

Advanced (Habanero)

Q1. What is the greatest common factor of 12, 18, and 24?

- (A) 1
- (B) 2
- (C) 3
- (D) 6
- (E) 12

Q2. Factor out the greatest common factor from $8x + 12y$

- (A) $2(4x + 6y)$
- (B) $4(2x + 3y)$
- (C) $4x + 6y$
- (D) $2x + 3y$
- (E) $4(2x + 3)$

Q3. Simplify the expression $9x^2 + 12x$ by factoring

- (A) $3x(3x + 4)$
- (B) $3x(3x + 2)$
- (C) $3x^2(3 + 4)$
- (D) $3x(3x - 4)$
- (E) $3x^2(4 + 3)$

Q4. Find the greatest common factor of 20, 30, and 40

- (A) 1
- (B) 2
- (C) 5
- (D) 10
- (E) 20

Q5. What is the factored form of $6x^2 + 8x$?

- (A) $2x(3x + 4)$
- (B) $2x(3x + 2)$
- (C) $2(3x^2 + 4)$
- (D) $2x(3x - 4)$
- (E) $2(3x^2 + 4x)$

Q6. Factor out the greatest common factor from $15x^3 + 25x^2$

- (A) $5x^2(3x + 5)$
- (B) $5x^2(3x + 2)$
- (C) $5x(3x^2 + 5)$
- (D) $5x^2(5x + 3)$
- (E) $5x^3(3x + 5)$

Q7. Simplify $16x + 20$ by factoring

- (A) $4(4x + 5)$
- (B) $4(4x + 2)$
- (C) $4(2x + 5)$
- (D) $4x + 5$
- (E) $2(8x + 10)$

Q8. Find the greatest common factor of 24, 36, and 48

- (A) 1
- (B) 2
- (C) 3
- (D) 6
- (E) 12

Open Ended Questions (FRQ)

Q9. Factor the expression $18x^2y + 24xy$ and simplify

Q10. Find the greatest common factor of 48, 60, and 72, and factor the expression $48x + 60y + 72z$

Chef's Tips

- Tip 1: Use the distributive property to check factored expressions
- Tip 2: Use prime factorization to find the greatest common factor
- Tip 3: Always simplify the expression after factoring